

Classical Mechanics

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Introduction

These are my notes on classical mechanics from **An Introduction to Mechanics** by **Daniel Kleppner and Robert J. Kolenkow**.

1 Vectors and Kinematics

Vectors were invented by **Willard Gibbs**. Vector makes it easy to write equations. For example, $F_x = ma_x, F_y = ma_y, F_z = ma_z$ can be written simply as $\vec{F} = m \cdot \vec{a}$. They are closely related to the fundamental ideas of symmetry (provides valuable insights). They can be defined geometrically as directed line segment.

Parallel translation doesn't change vector and same length and direction means equal vectors.

1.1 Dot Product

dot product or **scalar product**. It is defined as $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos \theta$ where θ is angle between two vectors.

Dot product can be thought as,

$$\vec{A} \cdot \vec{B} = |\vec{A}| \times (\text{projection of } \vec{B} \text{ on } \vec{A})$$

$$\vec{A} \cdot \vec{B} = |\vec{B}| \times (\text{projection of } \vec{A} \text{ on } \vec{B})$$

If dot product is zero, vectors are perpendicular to each other.

1.1.1 Law of Cosines

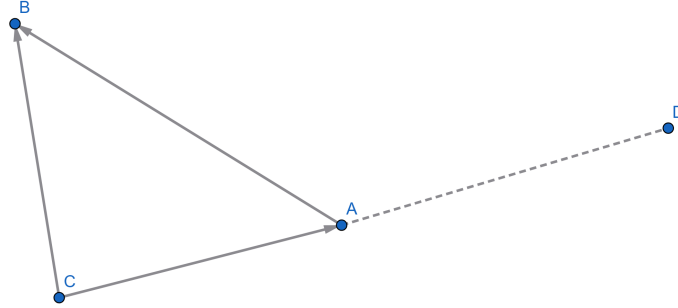


Figure 1: Law of Cosines

Let the angle $\angle BAC = \phi$ and $\angle BAD = \theta$. $\overrightarrow{CA} = \vec{a}$, $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{CB} = \vec{c}$.

$$\begin{aligned}\vec{c} \cdot \vec{c} &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ |\vec{c}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} \\ |\vec{c}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}| \cos \theta \\ |\vec{c}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}| \cos \phi\end{aligned}$$

1.1.2 Work(W)

$$W = F(\cos \theta)d = \vec{F} \cdot \vec{d}$$

in this definition of work, $F \cos \theta$ is the component of force along the direction of $\vec{d}$.

1.2 Vector Product

Vector product of two vectors is defined as

$$\vec{c} = \vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n}$$

here $\hat{n}$ is the unit vector in the direction perpendicular to both $\vec{a}$ and $\vec{b}$ which is determined by the right hand rule. Right hand rule says point all four fingers

of the right hand in the direction of first vector and curl them in the direction of the second vector, the direction in which the thumb points is the direction of $\hat{n}$. The coordinate system in which the z -axis point in the direction of the cross product from x -axis to y -axis is called **Right Handed Coordinate System**.

$$\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$$

$$\vec{a} \times \vec{a} = \vec{0}$$

1.2.1 Force experienced by charge

$$\vec{F} = q(\vec{v} \times \vec{B})$$

q is charge, $\vec{v}$ is velocity and $\vec{B}$ is magnetic field.

1.2.2 Torque

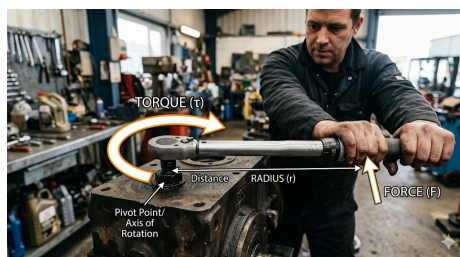


Figure 2: Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

1.2.3 Area as vector

This concept is useful in finding flow of water through wire loop (where orientation matters)

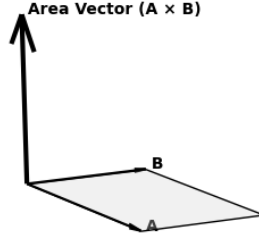


Figure 3: Area as vector

Area of parallelogram = $AB \sin \theta$. Thus, area vector has magnitude equal to that of cross product of two vectors and direction perpendicular to the plane of parallelogram.

1.3 Components of a Vector

Vector operations are defined without reference to coordinate systems.

1.3.1 2D Representation

For a vector $\vec{A}$ in a 2D plane:

- **Magnitude:** $|\vec{A}| = (A_x^2 + A_y^2)^{1/2}$
- **Direction:** $\tan \theta = \frac{A_y}{A_x}$
- **Notation:** $\vec{A} = (A_x, A_y)$

1.3.2 3D Representation

In 3D space, the vector is represented as:

$$\vec{A} = (A_x, A_y, A_z)$$

1.3.3 Proof: Magnitude in 3D

To find the magnitude $|\vec{A}| = (A_x^2 + A_y^2 + A_z^2)^{1/2}$, we use the Pythagorean theorem in two steps:

1. Find the projection of the vector in the xy -plane:

$$\text{Base length} = (A_x^2 + A_y^2)^{1/2}$$

2. Use this base and the vertical component A_z to find the final magnitude:

$$|\vec{A}| = \left[\left((A_x^2 + A_y^2)^{1/2} \right)^2 + A_z^2 \right]^{1/2}$$

$$|\vec{A}| = (A_x^2 + A_y^2 + A_z^2)^{1/2}$$

1.4 Coordinate Independence and Vector Algebra

$\vec{A}$ has meaning independent of any coordinate system; however, the components of $\vec{A}$ depend on the coordinate system chosen.

Coordinate System Comparison

Consider a vector $\vec{A}$ of magnitude A :

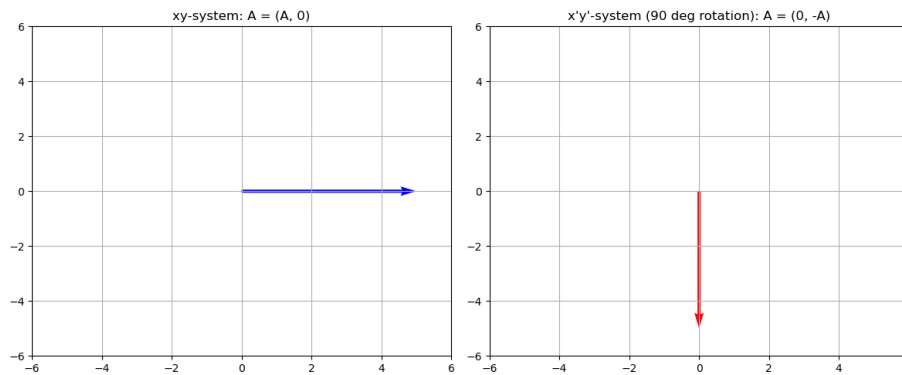


Figure 4: Coordinate Systems

- In the xy -system: $\vec{A} = (A, 0)$
- In the $x'y'$ -system: $\vec{A} = (0, -A)$ (rotated/shifted)

Vector Operations

To perform calculations, we typically restrict ourselves to a **single coordinate system**.

Vector Equality

If $\vec{A} = \vec{B}$, then their components must be identical:

$$A_x = B_x, \quad A_y = B_y, \quad A_z = B_z$$

Vector Addition

The sum of two vectors is the sum of their corresponding components:

$$\vec{A} + \vec{B} = (A_x + B_x, A_y + B_y, A_z + B_z)$$

Dot Product (Scalar Product)

The dot product is defined as the sum of the products of the components:

$$\vec{A} \cdot \vec{B} = (A_x B_x + A_y B_y + A_z B_z)$$

Base Vectors

A set of orthogonal unit vectors.

Properties of Unit Vectors

- **Dot Products:** $\hat{i} \cdot \hat{i} = 1$ and $\hat{i} \cdot \hat{j} = 0$
- **Cross Products:** $\hat{i} \times \hat{j} = \hat{k}$
- **Vector Representation:** $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$
- **Component Projection:** $A_z = \vec{A} \cdot \hat{k}$

Cross Product Derivation

The cross product $\vec{A} \times \vec{B}$ is calculated by distributing the terms:

$$\vec{A} \times \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

1st term expansion:

$$\begin{aligned} A_x \hat{i} \times \vec{B} &= A_x B_x (\hat{i} \times \hat{i}) + A_x B_y (\hat{i} \times \hat{j}) + A_x B_z (\hat{i} \times \hat{k}) \\ &= A_x B_x (0) + A_x B_y \hat{k} + (-A_x B_z) \hat{j} \\ &= A_x (B_y \hat{k} - B_z \hat{j}) \end{aligned}$$

Similarly:

$$\begin{aligned} A_y \hat{j} \times \vec{B} &= A_y (B_z \hat{i} - B_x \hat{k}) \\ A_z \hat{k} \times \vec{B} &= A_z (B_x \hat{j} - B_y \hat{i}) \end{aligned}$$

Determinant Form

Combining all terms, the cross product can be written as:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

1.5 Displacement and Position Vector

Key Principles

- **Vectors** have physical dimensions (e.g., meters, m/s).
- **Unit vectors** are dimensionless.

Definitions

- (a) **Position Vector ($\vec{r}$)**: Locates a point $P(x, y, z)$ relative to the origin O .

$$\vec{r} = (x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$$

Note: The position vector depends on the choice of the coordinate system.

- (b) **Displacement Vector ($\vec{S}$)**: Connects two points (x_1, y_1, z_1) and (x_2, y_2, z_2) .

$$\vec{S} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

Note: While the coordinates depend on the system, the displacement vector $\vec{S}$ itself does not depend on the coordinate system.

Coordinate Shift Invariance

If we have a shifted coordinate system where $\vec{R}$ is the shift vector:

$$\vec{R} + \vec{r}' = \vec{r} \implies \vec{r}' = \vec{r} - \vec{R}$$

For a displacement $\vec{S}$ between two points $\vec{r}_1$ and $\vec{r}_2$:

$$\vec{S} = \vec{r}_2 - \vec{r}_1$$

In the shifted system (r'):

$$\vec{S} = (\vec{R} + \vec{r}_2') - (\vec{R} + \vec{r}_1')$$

$$\vec{S} = \vec{r}_2' - \vec{r}_1'$$

This proves that the displacement vector remains identical regardless of the origin shift $\vec{R}$.

1.5 Velocity & Acceleration

Motion in 1D

- Average Velocity ($\bar{v}$):

$$\bar{v} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}$$

- Instantaneous Velocity (v):

$$v = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{dx}{dt}$$

- Acceleration (a):

$$a = \lim_{\Delta t \rightarrow 0} \frac{v(t + \Delta t) - v(t)}{\Delta t} = \frac{dv}{dt}$$

- Speed (s):

$$\text{speed } (s) = |\vec{v}|$$

Motion in Several Dimensions

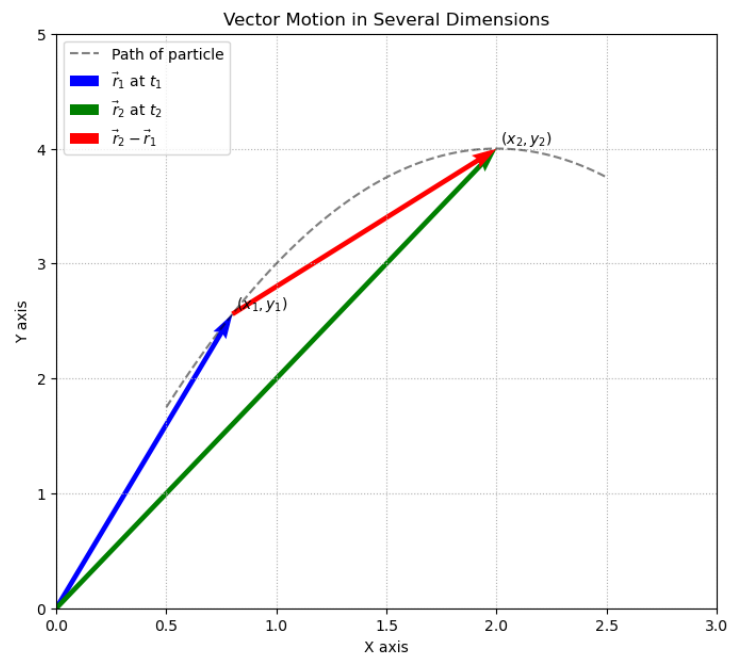


Figure 5: Motion in 2D

As shown in the coordinate system, $\vec{r}$ is a function of time.

- At t_1 time, position is (x_1, y_1) with radius vector $\vec{r}_1$.
- At t_2 time, position is (x_2, y_2) with radius vector $\vec{r}_2$.
- Displacement vector is given by $\vec{r}_2 - \vec{r}_1$.

Vector Displacement and Components

x_1 is the value of x at $t = t_1$, and at t_2 time, $\vec{r}_2 = (x_2, y_2)$. The displacement between times t_1 and t_2 is:

$$\vec{r}_2 - \vec{r}_1 = (x_2 - x_1, y_2 - y_1)$$

Vector vs. Scalar Components

In time Δt , the vector displacement is:

$$\Delta \vec{r} = \vec{r}(t + \Delta t) - \vec{r}(t)$$

This vector equation is equivalent to the scalar equations:

$\Delta x = x(t + \Delta t) - x(t)$ $\Delta y = y(t + \Delta t) - y(t)$

Velocity in 3 Dimensions

The instantaneous velocity vector is the limit of the displacement over time:

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

This is equivalent to the individual component velocities:

- **X-component:** $v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$
- **Y-component:** $v_y = \lim_{\Delta t \rightarrow 0} \frac{\Delta y}{\Delta t} = \frac{dy}{dt}$
- **Z-component (For 3D):** $v_z = \lim_{\Delta t \rightarrow 0} \frac{z(t+\Delta t) - z(t)}{\Delta t} = \frac{dz}{dt}$

$\vec{v} = \frac{d\vec{r}}{dt}$

Velocity as a Derivative of Position

The position vector $\vec{r}$ is defined in Cartesian coordinates as:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

The rate of change of the position vector with respect to time t gives the velocity vector:

$$\frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

The Geometric Interpretation

Given time intervals $\Delta t' < \Delta t'' < \Delta t'''$, we observe the relationship between displacement and the derivative:

- Approximation: $\Delta\vec{r} \approx \frac{d\vec{r}}{dt}\Delta t$
- Definition: $\Delta\vec{r} = \vec{v}\Delta t$

Note: This relationship becomes exact as $\Delta t \rightarrow 0$.

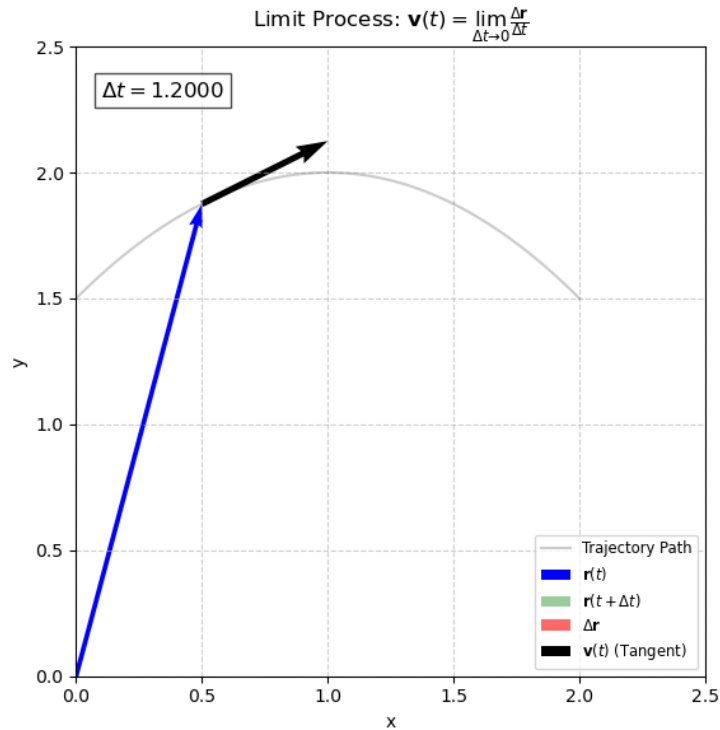


Figure 6: Velocity Vector

Conclusion

$$\boxed{\vec{v} \parallel \Delta \vec{r}}$$

As Δt approaches zero, the displacement vector $\Delta \vec{r}$ aligns with the direction of motion. Therefore:

$\vec{v}$ is tangent to the trajectory.

Kinematics of an Exponential Trajectory

[click here for animation](#)

Given the position vector $\vec{r}$ of a particle as a function of time t :

$$\vec{r} = A(e^{\alpha t} \hat{i} + e^{-\alpha t} \hat{j})$$

where α is a constant.

1. Velocity Vector

The velocity $\vec{v}$ is the first derivative of position with respect to time:

$$\vec{v} = \frac{d\vec{r}}{dt} = A(\alpha e^{\alpha t} \hat{i} - \alpha e^{-\alpha t} \hat{j})$$

The individual components are:

$$\begin{aligned} v_x &= A\alpha e^{\alpha t} \\ v_y &= -A\alpha e^{-\alpha t} \end{aligned}$$

The magnitude of velocity (speed) is:

$$|\vec{v}| = A\alpha \sqrt{e^{2\alpha t} + e^{-2\alpha t}}$$

2. Initial Conditions (at $t = 0$)

- Position: $\vec{r}(0) = A(\hat{i} + \hat{j})$
- Velocity: $\vec{v}(0) = \alpha A(\hat{i} - \hat{j})$

3. Asymptotic Behavior

As $t \rightarrow \infty$:

- $e^{\alpha t} \rightarrow \infty$ and $e^{-\alpha t} \rightarrow 0$.
- $\vec{r} \rightarrow Ae^{\alpha t} \hat{i}$ (The particle moves increasingly along the x -axis).
- $\vec{v} \rightarrow \alpha Ae^{\alpha t} \hat{i}$ (Speed increases without limit).

4. Acceleration Vector

The acceleration $\vec{a}$ is the derivative of velocity:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = A\alpha^2(e^{\alpha t}\hat{i} + e^{-\alpha t}\hat{j}) = \alpha^2\vec{r}$$

This indicates the acceleration is always proportional to the position vector.

Uniform Circular Motion

Uniform circular motion is defined as circular motion at a **constant speed**.
[click here for animation](#)

1. Position Vector

The position $\vec{r}$ of a particle in a circle of radius r is:

$$\vec{r} = r(\cos \omega t \hat{i} + \sin \omega t \hat{j})$$

Where:

- ω is the **angular velocity** (measured in radians/second).
- $|\vec{r}| = r$ (constant magnitude).
- At $t = 0$, $\vec{r}(0) = r\hat{i}$.
- Period T is the time for one complete revolution: $T = \frac{2\pi}{\omega}$.
- Relationship: $\omega T = 2\pi$.

2. Velocity Vector

The velocity $\vec{v}$ is obtained by differentiating position:

$$\vec{v} = \frac{d\vec{r}}{dt} = r\omega(-\sin \omega t \hat{i} + \cos \omega t \hat{j})$$

The speed is constant:

$$|\vec{v}| = r\omega = \text{constant}$$

Proof of Orthogonality: $\vec{v} \cdot \vec{r} = r^2\omega(-\sin \omega t \cos \omega t + \sin \omega t \cos \omega t) = 0$. This confirms $\vec{v} \perp \vec{r}$.

3. Centripetal Acceleration

The acceleration $\vec{a}$ is the derivative of velocity:

$$\vec{a} = \frac{d\vec{v}}{dt} = r\omega^2(-\cos\omega t\hat{i} - \sin\omega t\hat{j})$$

Simplified in terms of the position vector:

$$\vec{a} = -\omega^2\vec{r}$$

This is known as **centripetal acceleration**, as it is directed **radially inwards** (opposite to $\vec{r}$).

Formal Solution of Kinematical Equations

The fundamental relationship between acceleration $\vec{a}(t)$ and velocity $\vec{v}(t)$ is:

$$\frac{d\vec{v}(t)}{dt} = \vec{a}(t)$$

Discrete Approximation

By dividing the time interval $(t_1 - t_0)$ into n equal parts where $\Delta t = \frac{t_1 - t_0}{n}$, we can approximate the final velocity $\vec{v}(t_1)$ as a sum of incremental changes:

$$\begin{aligned}\vec{v}(t_1) &\approx \vec{v}(t_0) + \Delta\vec{v}(t_0 + \Delta t) + \Delta\vec{v}(t_0 + 2\Delta t) + \cdots + \Delta\vec{v}(t_1) \\ \vec{v}(t_1) &\approx \vec{v}(t_0) + \sum_{i=1}^n \vec{a}(t_0 + i\Delta t)\Delta t\end{aligned}$$

Where we use the approximation $\Delta\vec{v}(t) \approx \vec{a}(t)\Delta t$.

The Integral Limit

As $n \rightarrow \infty$ and $\Delta t \rightarrow 0$, the sum becomes a definite integral:

$$\vec{v}(t_1) = \vec{v}(t_0) + \int_{t_0}^{t_1} \vec{a}(t)dt$$

General Form

For an arbitrary time t , the velocity vector is given by:

$$\vec{v}(t) = \vec{v}_0 + \int_{t_0}^t \vec{a}(t')dt'$$

Similarly, the position vector $\vec{r}(t)$ can be found by integrating the velocity:

$$\vec{r}(t) = \vec{r}_0 + \int_{t_0}^t \vec{v}(t')dt'$$

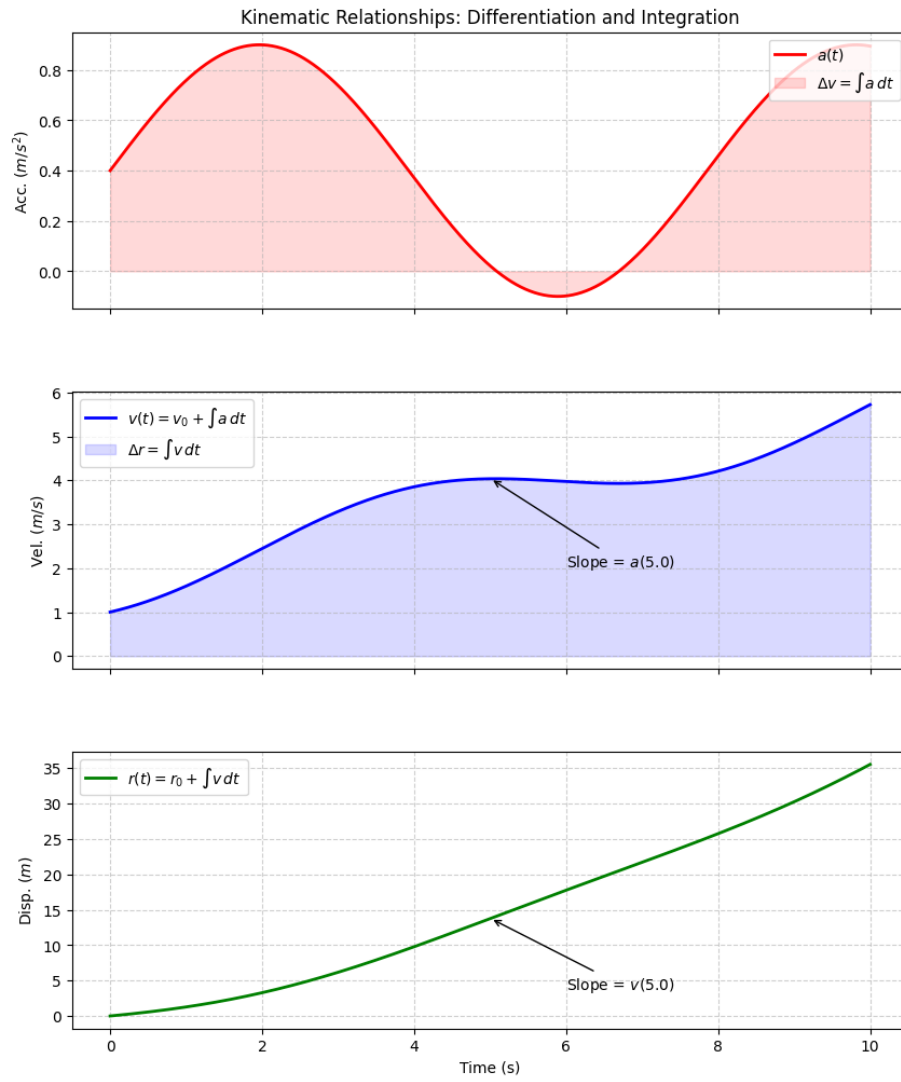


Figure 7: Graphs of Kinematical Equations

1.7 Formal Solution of Kinematical Equations

The fundamental relationship between acceleration $\vec{a}(t)$ and velocity $\vec{v}(t)$ is:

$$\frac{d\vec{v}(t)}{dt} = \vec{a}(t) \implies d\vec{v} = \vec{a}(t)dt$$

Integral Form

Integrating both sides from t_0 to t_1 :

$$\int_{t_0}^{t_1} d\vec{v} = \int_{t_0}^{t_1} \vec{a}(t)dt \implies \vec{v}(t_1) - \vec{v}(t_0) = \int_{t_0}^{t_1} \vec{a}(t)dt$$

For an arbitrary time t , the velocity vector is:

$$\vec{v}(t) = \vec{v}_0 + \int_{t_0}^t \vec{a}(t')dt'$$

Special Case: Uniform Acceleration

If $\vec{a}(t) = \vec{a} = \text{constant}$ and $t_0 = 0$:

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

To find the position $\vec{r}(t)$, we integrate the velocity:

$$\vec{r}(t) = \vec{r}_0 + \int_0^t (\vec{v}_0 + \vec{a}t')dt'$$

Which results in the standard kinematic equation:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2$$

Motion in a Uniform Gravitational Field

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In a uniform gravitational field near the Earth's surface, the acceleration vector $\vec{a}$ is constant and directed downwards:

$$\vec{a} = -g\hat{k}$$

1. Velocity and Position Vectors

Integrating the acceleration with initial conditions at $t_0 = 0$:

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2$$

2. Component Equations

Setting $\vec{r}_0 = 0$ (starting at the origin) and assuming motion in the xz -plane ($v_{0y} = 0$):

$$x = v_{0x}t \tag{1}$$

$$z = v_{0z}t - \frac{1}{2}gt^2 \tag{2}$$

3. Equation of the Trajectory

By eliminating t ($t = \frac{x}{v_{0x}}$), we obtain the equation of a parabola:

$$z = \left(\frac{v_{0z}}{v_{0x}} \right) x - \frac{g}{2v_{0x}^2} x^2$$

If the initial velocity $\vec{v}_0$ makes an angle θ with the x -axis:

- $v_{0x} = v_0 \cos \theta$
- $v_{0z} = v_0 \sin \theta$

The trajectory becomes:

$$z = x \tan \theta - \frac{gx^2}{2v_0^2 \cos^2 \theta}$$

Non-uniform Acceleration — Effect of Radio Wave on an Ionospheric Electron

[click here for animation](#)

Ionosphere: A region ~ 200 km above Earth's surface containing positively charged ions and negatively charged electrons.

When an oscillating electric field $\vec{E}$ interacts with these charges, it causes them to oscillate (or "jiggle").

Problem Statement

Find the motion of an electron (e^-) initially at rest, subjected to an oscillating electric field:

$$\vec{E} = \vec{E}_0 \sin(\omega t)$$

where ω is the angular frequency.

Derivation

By Newton's Second Law:

$$\vec{F} = m\vec{a} \implies -e\vec{E} = m\vec{a}$$

$$\vec{a} = \frac{-e}{m} \vec{E} = \frac{-e}{m} \vec{E}_0 \sin(\omega t)$$

Let $a_0 = \frac{-eE_0}{m}$. Then the acceleration is:

$$a(t) = a_0 \sin(\omega t)$$

1. Velocity $v(t)$:

$$v(t) = v_0 + \int_0^t a(t') dt' = v_0 + \int_0^t a_0 \sin(\omega t') dt'$$

$$v(t) = v_0 + a_0 \left[\frac{-\cos(\omega t)}{\omega} \right]_0^t = v_0 + a_0 \left(\frac{-\cos(\omega t) + 1}{\omega} \right)$$

$$v(t) = v_0 - \frac{a_0}{\omega} [\cos(\omega t) - 1]$$

2. Position $x(t)$: Assuming $x_0 = 0$ and $v_0 = 0$:

$$x(t) = \int_0^t v(t') dt' = \int_0^t \left(-\frac{a_0}{\omega} (\cos(\omega t') - 1) \right) dt'$$

$$x(t) = -\frac{a_0}{\omega} \left[\frac{\sin(\omega t')}{\omega} - t' \right]_0^t$$

$$x(t) = -\frac{a_0}{\omega} \left(\frac{\sin(\omega t)}{\omega} - t \right)$$

Final expression for position:

$$x(t) = \frac{a_0 t}{\omega} - \frac{a_0 \sin(\omega t)}{\omega^2}$$